

Air-Tracker: Flight Analytics

Project Report

Submitted By

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Abstract

Air-Tracker: Flight Analytics is a web-based flight analytics dashboard developed using Python, Streamlit, SQLite, Pandas, and Plotly. The application provides real-time-style flight analytics, airport information management, flight status monitoring, route analysis, and airport delay analysis through an interactive dashboard. The system helps users visualize and analyze aviation-related data efficiently using charts, tables, and search functionalities.

Objectives

- Develop an interactive flight analytics dashboard.
- Store and manage aviation data using SQLite.
- Provide flight search functionality.
- Analyze flight status distribution.
- Identify top flight routes.
- Analyze airport delay trends.
- Visualize aviation data through charts and tables.

Technologies Used

- Python 3
- Streamlit
- SQLite
- Pandas
- Plotly
- SQL

System Architecture

1. Database Layer (SQLite)
2. Data Processing Layer (Python & Pandas)
3. Visualization Layer (Plotly)
4. User Interface Layer (Streamlit)

Features Implemented

1. Dashboard Overview

Displays:

- Total Airports
- Total Aircraft
- Total Flights
- Total Delay Records

2. Flight Search

Users can search flights using flight numbers and view:

- Flight Number
- Airline Code
- Origin Airport
- Destination Airport
- Flight Status

3. Airport Information

Displays airport details including:

- Airport Code
- ICAO Code
- Airport Name
- City
- Country

4. Flight Status Analysis

Visual representation of:

- Boarding Flights
- Delayed Flights
- Departed Flights
- Gate Closed Flights
- Expected Flights
- Cancelled Flights

5. Top Routes Analysis

Displays the most frequently operated routes.

6. Airport Delay Analysis

Shows:

- Average Delay
- Arrival Delay
- Departure Delay for major airports.

Database Design

Airports Table

- airport_id
- iata_code
- icao_code
- airport_name
- city
- country

Flights Table

- flight_id
- flight_number
- airline_code
- origin_iata
- destination_iata
- status

Delay Table

- airport_iata
- average_delay
- arrival_delay
- departure_delay

Results

The dashboard successfully provides:

- Interactive flight analytics
- Flight status monitoring
- Route analysis
- Airport delay visualization
- Airport information management

Future Enhancements

- Real-time Flight API Integration
- User Authentication

- Flight Booking Module
- Predictive Delay Analytics using Machine Learning
- Cloud Deployment (AWS/Azure)

Conclusion

Air-Tracker: Flight Analytics successfully demonstrates the use of Python, Streamlit, SQLite, and data visualization tools to create an interactive aviation analytics platform. The project provides meaningful insights into flight operations, airport performance, and route trends while offering a user-friendly dashboard for data exploration.

GitHub Repository

Air-Tracker-Flight-Analytics

Developed By

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